

TABLE II
INSTRUCTION SETS

Opcode	Instruction	Data	Description
0001	LDI	<immediate data>	Load the immediate value to the accumulator
0010	NOT	—	Perform NOT operation on the value stored in the accumulator and store the result in the accumulator
0011	AND	<immediate data>	Perform AND operation on the value stored in the accumulator and the immediate value, and store the result in the accumulator
0100	OR	<immediate data>	Perform OR operation on the value stored in the accumulator and the immediate value, and store the result in the accumulator
0101	XOR	<immediate data>	Perform XOR operation on the value stored in the accumulator and the immediate value, and store the result in the accumulator
0110	ADD	<immediate data>	Perform ADD operation on the value stored in the accumulator and the immediate value, and store the result in the accumulator
0111	SUB	<immediate data>	Perform SUB operation on the value stored in the accumulator and the immediate value, and store the result in the accumulator
1000	NOP	—	No operation, halt CPU
1001	LDA	<immediate address>	Load the value stored in the address location to the accumulator
1010	STO	<immediate address>	Store the accumulator value to the address location specified in the immediate value
1011	JPE	<immediate address>	Jump the PC to memory location 'immediate+1' if the accumulator value is equal to B-register's value
1100	JPG	<immediate address>	Jump the PC to memory location 'immediate+1' if the accumulator value is greater than of B-register
1101	JPL	<immediate address>	Jump the PC to memory location 'immediate+1' if the value of the accumulator is less than of B-register
1110	OUT	—	Output value is stored in the accumulator
1111	B-REG	<immediate data>	Store the immediate value in B-register

Programming

Programming this computer is simple. There are two on/off switches (S8 for clock on/off and S7 for program/run mode selection), four pushbutton switches (S3 through S6) and two 8-bit DIP switches (one for address and the other for data). S3, S5 and S6 are reset switches, while S4 is RAM-write switch.

Toggle switch S7 to 'program' mode. The LED connected to it should turn off. Switch the clock to 'off' mode. Switch 8-bit address to the first address (0b00000001) and set the data DIP switch to the instruction/data to be written on the address. Press RAM-write switch

S4 to load data onto the address. Similarly, repeat the process for successive code lines.

After entering all the codes/inputs, toggle program/run switch S7 to 'run' mode. Reset switches S3, S5 and S6 to clear all the LEDs. Press switch S8 on the clock to start computation. The maximum number of code lines that can be written for this project is 255.

Various instruction sets are listed in Table II.

Example. The following code loads 45 in the accumulator, performs NOT operation on it, ADDs 34 to the result, displays the output and halts:

```
LDI 45
NOT
```

TABLE III
SEQUENCE OF OPERATION

Address	Data	Code	Remarks
00000001	00000001	LDI	Instruction
00000010	00101101	45	Data
00000011	00000010	NOT	Instruction
00000100	00000010	ADD	Instruction
00000101	00100010	34	Data
00000110	00001110	OUT	Instruction
00000111	1000	NOP	Instruction

```
ADD 34
OUT
NOP
```

The sequence of operation is as follows:

1. Stop the clock mode
2. Clear all resets (press reset 1→2→3→1 or press S6→S3→S5→S6)
3. Switch S7 to 'program' mode
4. Set address to 00000001 using DIP switch
5. Set data DIP switch to value 00000001
6. Press RAM-write button S4
7. Set address DIP switch to next address 00000010
8. Set data DIP switch to value 00101101 (binary for 45)
9. Press RAM-write switch S4
10. Repeat it until the NOP (00001000) is set in the RAM
11. Switch S7 to 'run' mode
12. Start the clock using switch S8

The sequence of operation is shown in Table III.

LDI instruction flow diagram (not shown here) is included in the EFY DVD of this month.

ROM code

The address from the instruction register, counter and comparator is fed to the two ROMs. The ROM code is provided in a separate file in the EFY DVD.

Burn the exact code into ROM1 and ROM2. For details on burning the ROM code, refer the datasheet of AT28C64. All the ROM address fields must be filled with the following values to ensure there is no garbage